

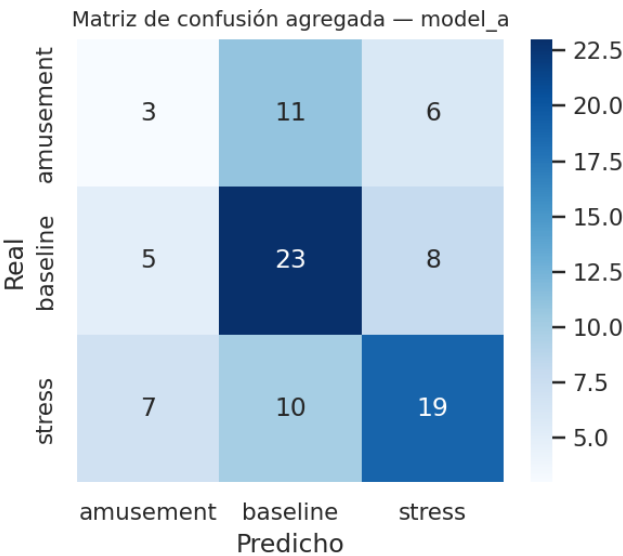
# Gemelo Digital de Evacuación Minera

Reporte automático del Motor IA — generado el 2026-09-27 17:16 UTC

## 2. Métricas por arquitectura (validación cruzada)

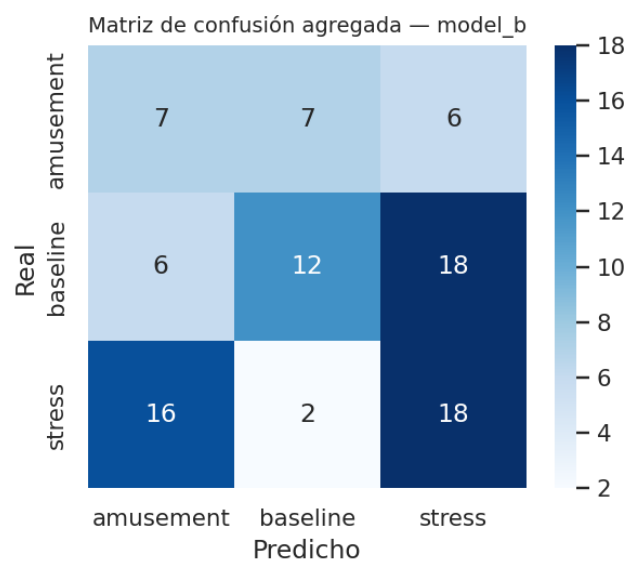
**model\_a** (4 folds)

| Métrica         | Media  | Desv. estándar |
|-----------------|--------|----------------|
| accuracy        | 0.4891 | 0.1205         |
| precision_macro | 0.3887 | 0.0867         |
| recall_macro    | 0.4389 | 0.0900         |
| f1_macro        | 0.3727 | 0.0863         |
| cohen_kappa     | 0.2023 | 0.1731         |
| auc             | 0.6299 | 0.1496         |



**model\_b** (4 folds)

| Métrica         | Media  | Desv. estándar |
|-----------------|--------|----------------|
| accuracy        | 0.4022 | 0.1609         |
| precision_macro | 0.3638 | 0.2073         |
| recall_macro    | 0.3944 | 0.1676         |
| f1_macro        | 0.3596 | 0.1646         |
| cohen_kappa     | 0.0962 | 0.2495         |
| auc             | 0.5951 | 0.1824         |



### 3. Comparación estadística de arquitecturas

Métrica comparada: **f1\_macro**. Medias: model\_a=0.3727, model\_b=0.3596

| Test                 | Método A | Método B | p-valor | Significativo ( $\alpha=0.05$ ) | Mejor |
|----------------------|----------|----------|---------|---------------------------------|-------|
| wilcoxon_signed_rank | model_a  | model_b  | 0.875   | No                              | —     |

**Decisión estadísticamente justificada:** empate\_no\_significativo